

The Graph Laplacian and the Continuous Laplace Operator

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Graph Laplacian Review

Initial Data

Datapoints $\{x_i\}$

Similarity measure $s(x_i, x_j)$

Build the Graph

Adjacency

$$A_{ij} = s(x_i, x_j)$$

Degree of x_i

$$D_{ii} = \sum_j s(x_i, x_j)$$

- Then we define the graph Laplacian:

$$L = D - A$$

Why Is It Called The Graph "Laplacian"?

- Namesake
 - The continuous Laplace operator is the sum of the second partial derivatives
$$\Delta f = - \sum_i \frac{\partial^2}{\partial x_i^2} f.$$
 - This operator "knows a lot about geometry": the heat equation $\partial_t u(t, x) = \Delta u(t, x)$ models how heat flows through the shape u is defined on.
- This is not a coincidence!
 - Belkin and Niyogi (2008) showed that the (appropriately scaled) graph Laplacian converges to the continuous Laplace operator as we sample more points.
 - An intuitive connection: $(Lv)_i = \sum_j A_{ij}(v_i - v_j)$ measures the difference between vertex i and its neighbors, weighted by similarity $A_{ij} = s(x_i, x_j)$; $(\Delta f)(x)$ measures how different $f(x)$ is from $f(y)$ for points y near x .

Some Background

- In Euclidean space we can compute the distance between points and angles between vectors with the dot product:

$$d(x, y) = \sqrt{\sum_i (x_i - y_i)^2} = \sqrt{(\vec{x} - \vec{y}) \cdot (\vec{x} - \vec{y})}$$

$$\cos(\theta) = \frac{\vec{x} \cdot \vec{y}}{\|\vec{x}\| \|\vec{y}\|} = \frac{\vec{x} \cdot \vec{y}}{\sqrt{(\vec{x} \cdot \vec{x})(\vec{y} \cdot \vec{y})}}.$$

- On a curved space, we have to be more careful:



Figure: Hyperboloid of one sheet

Using The Graph Laplacian To Access The Metric

- A calculation gives the product rule for the Laplacian:

$$\Delta(fg) = f \Delta g + g \Delta f - 2 \langle \nabla f, \nabla g \rangle_M,$$

or $\langle \nabla f, \nabla g \rangle_M = \frac{1}{2}(f \Delta g + g \Delta f - \Delta(fg)).$

- So we can write $\langle \nabla f, \nabla g \rangle_M$ in terms of the Laplace operator Δ , which we know how to approximate with the graph Laplacian.
- Since this inner product $\langle \cdot, \cdot \rangle_M$ knows how the manifold curves, it contains all the information about the geometry of M .

Results: Pointwise Dimension

We compute the matrix $\hat{G}$, a representation of $\langle \cdot, \cdot \rangle_M$. Then looking at its eigenvalues tells us the dimension near that point.

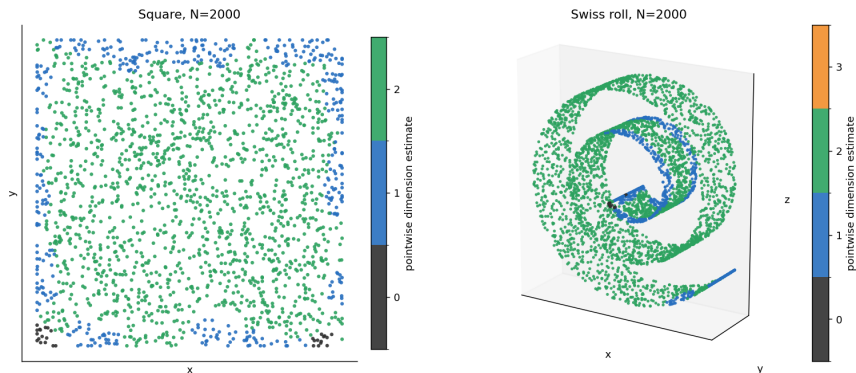


Figure: Dimension estimate

Results: Tangent Spaces

Diagonalizing $\hat{G}$ gives us a basis for the tangent space, i.e. the set of directions we can move in our data. We use synthetic data sampled from a torus and hyperboloid of one sheet.

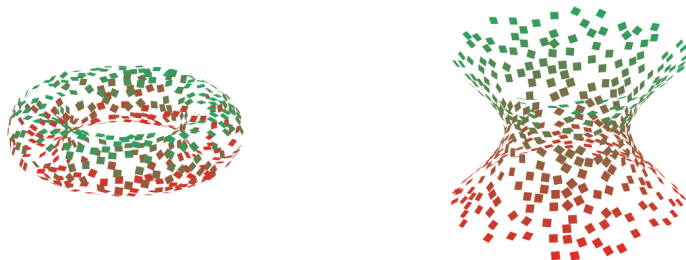


Figure: Approximated tangent planes

Results: Added Noise

We add noise $\vec{x} \sim \mathcal{N}(0, \sigma^2)$ to the synthetic data for the values of σ on the x-axis.

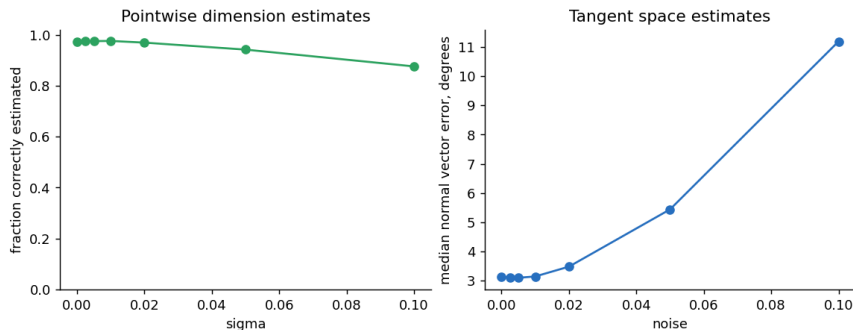


Figure: Error in results with added noise